Western Governor's University

D210 Reporting & Visualization

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A. In a document file, do the following:

1. Provide a copy of your interactive Tableau dashboard to support executive decision-making. Your dashboard must be accessible to users with colorblindness and must include the components in each of the following bullet points:

See file attached: “Brandon Velez D210v2.twb”

- data integrated from two sources: one of the provided data sets and one additional external, public data set
- four different data representations to summarize the data or display trends
- two different interactive controls that allow the user to modify the presentation of data
- two different metrics or key performance indicators computed using data from both chosen data sets

2. Provide both data sets that serve as the data source for the dashboard.

See files attached: medical_clean.csv, epa_air_quality_annual_summaryBVD210.csv

(Kaggle,2019)

3. Provide step-by-step instructions to guide users through the dashboard installation.

Download dataset csv files onto computer. Open tableau>Data>New data source> connect to Text file. Then navigate to the csv file and select. Repeat for other data set.

After downloading the workbook, open Tableau>Navigate to the top left corner to find File>Open> and locate and select “Brandon Velez D210.twb”.

4. Provide instructions to help users navigate the dashboard.

After opening the workbook through tableau, select “Present Story”. Enter presentation mode using fn+F7 keys or selecting the presentation icon on the far right of the actions toolbar just above the graphs presented. Now, let’s look at the “Introduction to Readmission Problem” pane. Here you can see a bar chart

visualizing readmission percentage by bins of initial days. You have the ability to scroll up and down and to hover over a row to get tooltip details. On the right side of this page, you will see filters to adjust the chart below, visualizing average pollutant measurement per year. These filters include pollutant name, state, and a slider for date range.

Now select the “pollutant and readmission by county maps” pane. Here there are two maps, the top visualizes the most prevalent pollutant per county, and below visualizes readmission rates by county. The year and pollutant filters on the right are for the pollutant map up top, along with the pollutant color legend. There is also a state filter which is applied to both maps. For the readmission percentage map, there is a slider filter to select data points by readmission percentage range, directly above a legend to clarify the color usage.

Select the ‘Texas KPIs’ pane. This visualizes the recently discovered Calcium PM10 pollutant in Texas by year, along Texas’ readmission rate.

Finally, select the “pollutants by state bar chart”. This is a bar chart allowing users to visualize pollutants by state. Filters on the right allow for selection of pollutants, state and county location, a slider for date range, and a legend to clarify color usage.

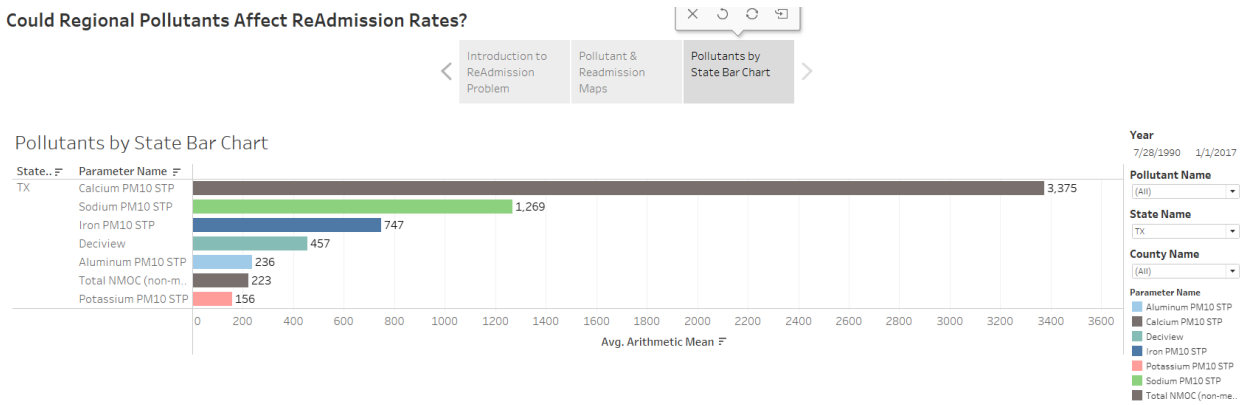
Part 2: Storytelling with Data

B.

Panopto Link: https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=cfe02a84-e3dc-4023-b631-b304014b68f5

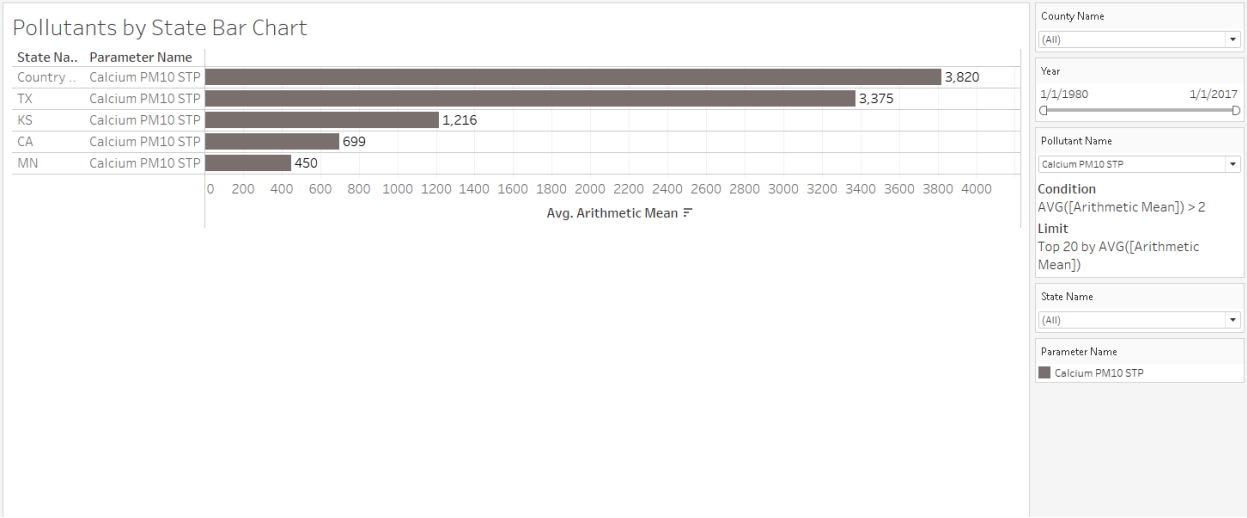
- an introduction of yourself and your background
- a summary of both chosen data sets and the context in which they occur
- an outline of key results from your analysis of the two data sets

It is clear patients who spend more time in the hospital shown by the initial days measurement are at higher risk of readmission. Presumably, these patients are often difficult to diagnose and treat, as they need the most time in the hospital and return at the highest rate despite receiving the most treatment.



- two different data representations to serve as supporting evidence for your results
- a summary of actionable insights based on your results

Hospitals with high readmission rates should perform an analysis using the dashboard controls to determine the greatest pollutants in their state/county and then research if those pollutants have been linked to health issues. For example, a hospital in Texas may benefit from testing patients for the pollutant Calcium PM10 STP. This is not only the largest pollutant by volume in the state of Texas, but Texas is outputting more than twice as much of this pollutant than any other state. Total hydrocarbons and methane emissions could also be further researched and tested for correlation with chronic health issues that cannot otherwise be attributed and are potentially causing readmission.



Note: Your two data representations may come directly from your Tableau dashboard created in part A.

Part 3: Reflection Paper

C. In a document file, write a reflection paper to demonstrate your understanding of data representation and reporting by doing the following:

1. Explain how the purpose and function of your dashboard align with the needs outlined in the data dictionary associated with your chosen data set.

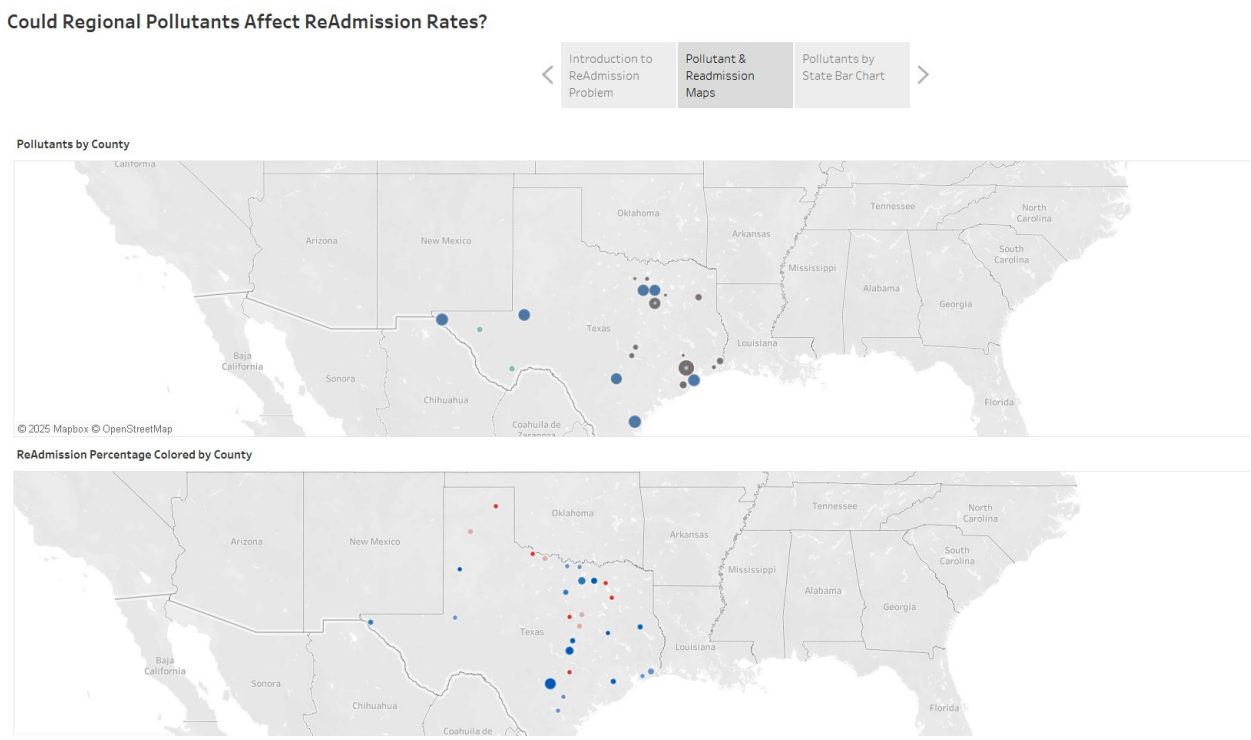
The medical data dictionary states that hospitals are struggling with patient re-admission rates being too high. As hospitals are unaware of what factors could be causing this, there may be useful insights in finding correlations between pollutants and nearby hospital re-admission rates. This dashboard will allow end users to build a map visualizing the highest hospital re-admission rates and volume, allowing them to visualize which pollutants are most common nearby. The pollutant by state bar chart allows deeper analysis, showing the 20 highest pollutants by arithmetic mean, with the option to group and filter by state or county. These data visualizations are meant to suggest an alternative perspective into potential chemical exposure/inhalation causing otherwise undiagnosable and untreatable readmissions.

2. Explain how the variables in the additional data set enhance the insights that can be drawn from the data set you chose from the provided options.

The pollutant data set will allow visualizations using state and county, which can be the linking field between these two data fields. This will allow for a worksheet showing highest pollutants by region to compare to a map of pollutants. The pollutant dataset includes measures of each pollutant, pollutant names, measurement units, location of measurement, and other details are included to allow for detailed geographic correlation and annual analysis.

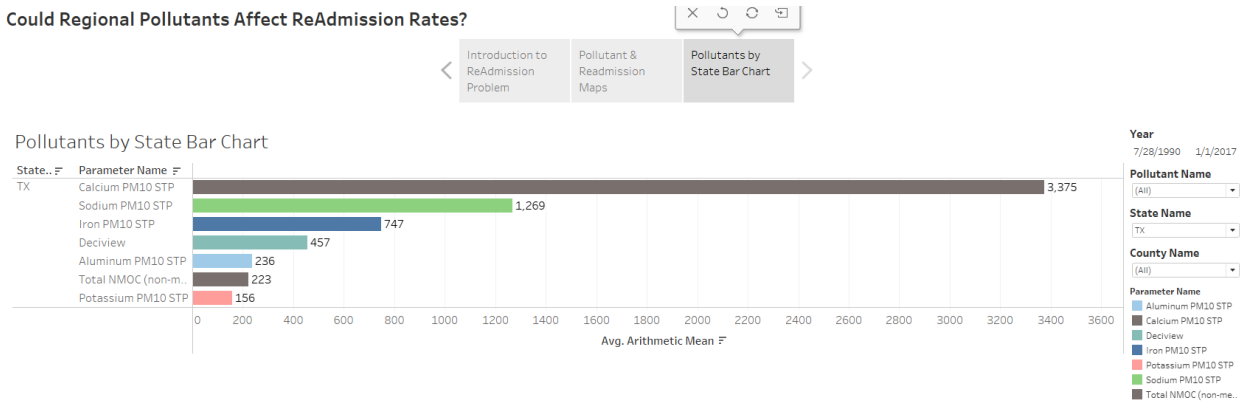
3. Explain two different data representations from your dashboard and how executive leaders can use them to support decision-making.

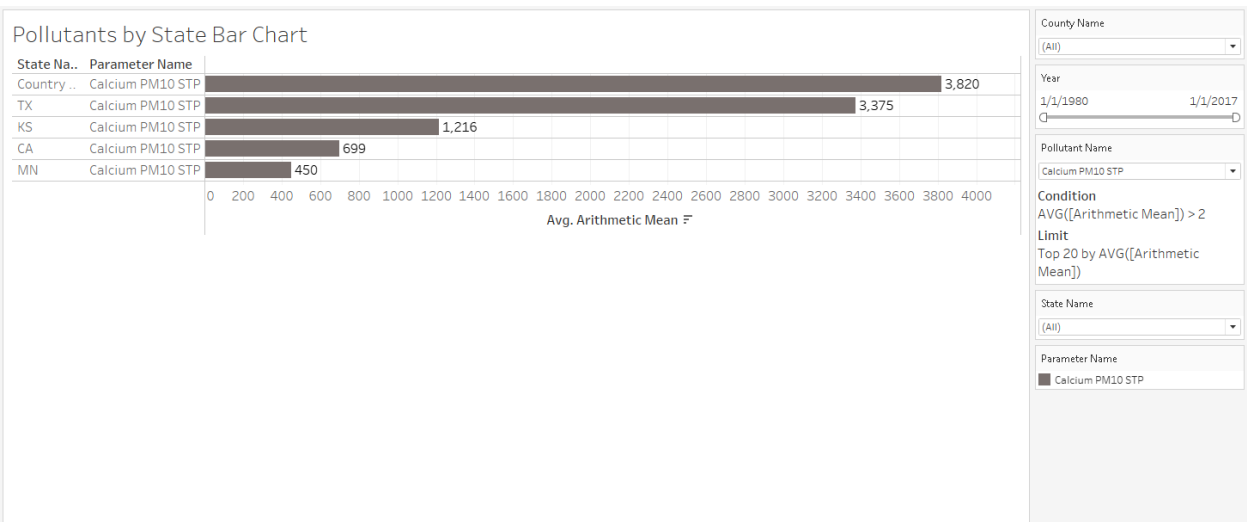
The county dashboard can be used by executives to assess the counties with the highest readmission rates in the country, filtering out counties with lower readmission rates for clarity. This can be compared to the map directly above, which shows the geographic distribution of pollutants. This allows a filter on parameter name to show however many pollutants at a time are desired. Finding the pollutants that stand out as having a noticeably higher distribution in the areas with the highest readmission rates could indicate that hospitals should begin testing for pollutant buildup in area patients.



The pollutants by state bar chart worksheet is set up for executives to analyze pollutant data by either selecting the pollutants they would like to see

distributed across states, or select state/county names to analyze which pollutants are most prevalent in a given area. This will populate a list of pollutants in chosen states and counties to suggest which pollutants to research or keep data on.





4. Explain two interactive controls in your dashboard and how each enables the user to modify the presentation of the data.

The pollutant measurements dashboard has both a filter for pollutant name and state name. This allows users to visualize which pollutants are most prevalent per state, and the ability to visualize which states include any selected pollutants.

The county dashboard includes a sliding filter to include readmission rates only within any chosen range which will visualize on the map immediately.

5. Describe how you built your dashboard to be accessible for individuals with colorblindness.

These visualizations have been made using colorblind safe color palettes. There are so many different pollutant names however, if they are all on the map or graph at once, it will still be difficult to distinguish between colors. This can be mitigated by using the shown pollutant name filter to view a smaller set of more visually distinguishable pollutants, or by viewing a zoomed in region, which is presumably most users' use case.

(Nichols, 2017)

6. Explain how two data representations in your presentation support the story you wanted to tell.

The data has been presented to show executives that pollutants have been significantly rising over time using the annual pollution bar chart, alongside the problematic issue of readmission rates being so high for patients with longer initial stays using the readmission rate by initial days visualization. We will visualize and discuss how these pollutants have grown without an understanding of future health detriments, and how these factors go largely ignored in health analyses.

Then, a more local area such as the state of Texas can be chosen. Readmission rates across the state can be shown by color on the pollution by county map, and then pollutants in the state can be listed using the pollutants by state bar chart.

7. Explain how you used audience analysis to adapt the message in your presentation.

The medical data dictionary outlines that this presentation will be shown to the Sr Vice President of Hospital operations, the VP of research, regional VPs, and data analytics peers. The executives are likely not well versed in building data visualizations, so plain language is used without explaining the complexity of the Tableau calculations. Presumably, the executives would like an efficient presentation that communicates a clear point without spending time on how the dashboard was built. That is why all relevant filters are already added and have

clarified, bolded titles with simple dropdowns, these are fully self-explanatory while capturing all use cases. Tooltip has also been clarified to give clear details such as unit of measure and exact measurements along with location data, while removing any information that is non-necessary. Additionally, the charts and maps have been created using a wide color range to distinguish pollutants and draw the audience in with some intriguing visuals.

8. Describe how you designed your presentation for universal access by all audiences.

The story and relating dashboards have been optimized for end users with no tableau experience or understanding of these datasets. Data fields have been renamed to clarify and simplify what each graph or map is visualizing, and filters have also been given clarified titles and clear legend visualizations so that users with no background knowledge will be able to intuitively visualize desired data from these sets. Where appropriate, color schemes have also been added as an additional layer of visualization with legends on the right pane. All irrelevant filters, fields, or text have been removed from the worksheets, dashboards, and story. Colors have been modified for accessibility by colorblind individuals as well.

9. Explain two elements of effective storytelling that you implemented in your presentation and how each element was intended to engage the audience.

Strong visualization is the most important element of storytelling in this presentation. Each map or chart has been carefully colored to ensure clear distinction between points while drawing in attention. Filters have been edited to be as clear and concise as possible while allowing users full customization. All text shown in the presentation has been minimized to keep the focus on the data, but necessary clarifications have been input.

An element this presentation relies on is the uncertainty around readmission rates and these pollutants, that can cause a bit of discomfort to hook the audience. Showing the audience that we don’t understand why patients are being readmitted and following it up with discussion of how little these measured pollutants are understood introduces a significant element of unknown that our brains very much want to resolve to break that sense of incompleteness. This

presentation is framed as a potential answer to some health issues – it suggests a correlation while allowing audience members to conclude if they find these pollutants a potential factor in readmission or not, encouraging the audience to explore the visualizations more deeply using the readily available filters.

D. Sources:

Works Cited

"Air Quality Annual Summary." *Kaggle.com*, 2017, www.kaggle.com/datasets/epa/air-quality. Accessed 2 June 2025.

Nichols, David. "Coloring for Colorblindness." *Davidmathlogic.com*, 2019, davidmathlogic.com/colorblind/#%23D81B60-%231E88E5-%23FFC107-%23004D40.